

Q1. DIRECTIONS *for question 1:* Type in your answer in the input box provided below the question.

A given sequence of numbers is such that in any set of four consecutive numbers, the sum of the first and third terms is equal to the sum of the second and fourth terms. If the 3rd and 14th terms are 4 and 7 respectively and the sum of the first 18 terms is 57, what is the sum of the first 23 terms?

Q2. DIRECTIONS *for questions 2 and 3:* Select the correct alternative from the given choices.

In triangle PQR, S is a point on QR such that PS bisects $\angle QPR$. If $\angle QPR = 45^\circ$ and $\frac{QS}{SR} = \sqrt{2}$, find $\angle PQR + \angle PSR$.

Q3. DIRECTIONS *for questions 2 and 3:* Select the correct alternative from the given choices.

If x satisfies the inequality $|x - 1| + |x - 2| + |x - 3| \leq 6$, then which of the following options best describes the range of values that x can assume?

☐ a) $x \leq 2$ or $x \geq 3$

☐ b) $x \leq 1$ or $x \geq 4$

☐ c) $0 \leq x \leq 4$

☐ d) $x \leq 0$ or $x \geq 4$

Q4. DIRECTIONS *for question 4:* Type in your answer in the input box provided below the question.

A communication tower of height 10 m is erected on the top of a building of height 30 m. If the distance from the base of the building to the point on the ground, at which the tower subtends the greatest angle is x m, find

$$\sqrt{3}x. \text{ Given } \tan (A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}.$$

Q5. DIRECTIONS *for question 5:* Select the correct alternative from the given choices.

A shopkeeper had 15 cartons of pencil boxes with him and each carton had 120 pencil boxes, with each pencil box having 15 pencils. If the shopkeeper was able to sell only 10% of the pencils present in 15% of the pencil boxes in 20% of the cartons that he had, how many pencils was the shopkeeper able to sell?

☐ a) 81

☐ b) 99

☐ c) 72

☐ d) 78

Q6. DIRECTIONS *for questions 6 and 7:* Type in your answer in the input box provided below the question.

A cuboid has a volume of 64 cubic units. Find the minimum possible value (in units) of the sum of the lengths of all the edges of the cuboid.

Q7. DIRECTIONS *for questions 6 and 7:* Type in your answer in the input box provided below the question.

ABC is a three-digit number such that $ABC = 5(AB + BC + CA)$, where AB , BC and CA are all two-digit numbers. Find the total number of possible values for the number ABC .

Q8. DIRECTIONS *for question 8:* Select the correct alternative from the given choices.

If the largest angle of a triangle is twice the sum of the other two angles, find the maximum possible ratio of the shortest side of the triangle to that of the longest side of the triangle.

Q9. DIRECTIONS *for question 9:* Type in your answer in the input box provided below the question.

In the X-Y coordinate system, the area (in sq.units) of the region bounded by the relation $|x| + |x + y| = 6$ is

Q10. DIRECTIONS *for questions 10 to 12:* Select the correct alternative from the given choices.

The standard deviation of the observations 4086, 4090, 4082, 4094, 4078, 4102, 4098 is

- ☐ a) $\sqrt{8}$
- ☐ b) 8.
- ☐ c) 8.25.
- ☐ d) 7.25.

Q11. DIRECTIONS *for questions 10 to 12:* Select the correct alternative from the given choices.

John participated in a 500 m race, during which he increased his speed by 10% after every 100 m that he covered. If he finished the race in 3 minutes and 14 seconds, what was his approximate speed for the first 100 m of the race?

- ☐ a) 7.73 kmph
- ☐ b) **6.12 kmph**
- ☐ c) 6.47 kmph
- ☐ d) 7.12 kmph

Q12. DIRECTIONS for questions 10 to 12: Select the correct alternative from the given choices.

Let x, y, z be three positive numbers, such that $x + y + z = 1$. The minimum possible value of $\left(x + \frac{1}{x}\right)^2 + \left(y + \frac{1}{y}\right)^2 + \left(z + \frac{1}{z}\right)^2$ is

☐ a) 27.

☐ b) **18.**

☐ c) $33\frac{1}{3}$

☐ d) 40.

Q13. DIRECTIONS *for questions 13 to 15:* Type in your answer in the input box provided below the question.

Five friends A, B, C, D, E went to a tiffin centre for breakfast. Of the five items available, A, B, C, D, E do not like idly, puri, dosa, upma, vada respectively. Each person orders exactly one item and between them, they order all the five items. In how many ways can they order their breakfast?

Q14. DIRECTIONS *for questions 13 to 15:* Type in your answer in the input box provided below the question.

In a country called Numsyland, the people use a certain number system to the base n ($n > 2$) and the currency of the country is Nollar. Franky, a resident of the country, one day received twice his daily wage because the digits of his wage (in Nollars), which was a two-digit number, were accidentally reversed. If the value of n is the least possible, then what is the decimal representation of the excess wage (in Nollars) paid to Franky on that day?

Q15. DIRECTIONS *for questions 13 to 15:* Type in your answer in the input box provided below the question.

If $(7 + 4\sqrt{3})^{20} = I + F$ and $(7 - 4\sqrt{3})^{20} = f$, where I is an integer, $0 < F < 1$ and $0 < f < 1$, what is the value of $42(F + f)$?

Q16. DIRECTIONS *for questions 16 to 19:* Select the correct alternative from the given choices.

A, B and C started a job. On the first day, C being unwell is not able to work at his full capacity and he leaves after one day. The other two complete the work and B gets a share of ₹5100 out of a total of ₹18000 paid for the job. If each of A, B and C can complete the job normally in 10 days, 20 days and 5 days respectively, then find the percentage efficiency with which C worked on the first day.

- ☐ a) 50%
- ☐ b) **60%**
- ☐ c) 75%
- ☐ d) 80%

Q17. DIRECTIONS *for questions 16 to 19:* Select the correct alternative from the given choices.

If a, b, c are single-digit positive integers, such that $\log_5 \log_4 \log_3 \log_2(a^{b^c}) = 0$, then find the maximum possible value of $a + b + c$.

- ☐ a) 9
- ☐ b) 13
- ☐ c) 17
- ☐ d) 14

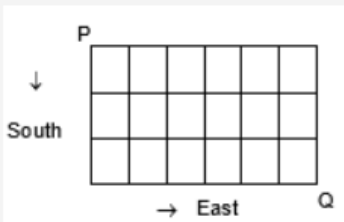
Q18. DIRECTIONS *for questions 16 to 19:* Select the correct alternative from the given choices.

Given $N = 35 \times 36 \times 37 \dots \times 67$, what is the remainder left when N is divided by 289?

- ☐ a) 3
- ☐ b) **238**
- ☐ c) 51
- ☐ d) 286

Q19. DIRECTIONS for questions 16 to 19: Select the correct alternative from the given choices.

In the figure below, the lines represent one-way roads which allow travel only southwards or only eastwards. Find the number of distinct routes by which a car can reach point Q from point P.



- ☐ a) 64
- ☐ b) **84**
- ☐ c) 56
- ☐ d) 90

Q20. DIRECTIONS *for questions 20 and 21:* Type in your answer in the input box provided below the question.

M and N are two natural numbers, such that the indices of the highest powers of 5 in N! and M! are 64 and 28 respectively. Find the maximum difference between the values of N and M.

Q21. DIRECTIONS *for questions 20 and 21:* Type in your answer in the input box provided below the question.

Find the area (in sq. units) of the polygon formed by joining the points $(2, 0)$, $(3, 2)$, $(4, 6)$, $(1, 7)$, $(-1, 4)$ and $(0, 2)$, in that order.

Q22. DIRECTIONS *for question 22:* Select the correct alternative from the given choices.

x is an integer such that $16 \leq x \leq 81$.

If $-y = \frac{x^2 + 3\sqrt{x}(2x + 9) + 162}{x + 9\sqrt{x} + 18}$, then what is the range of y ?

- ☐ a) $-28 \leq y \leq 75$
- ☒ b) $39 \leq y \leq 52$
- ☐ c) $13 \leq y \leq 63$
- ☐ d) $-63 \leq y \leq -13$